

Flask + Redis + Nginx: Containerized Cloud App

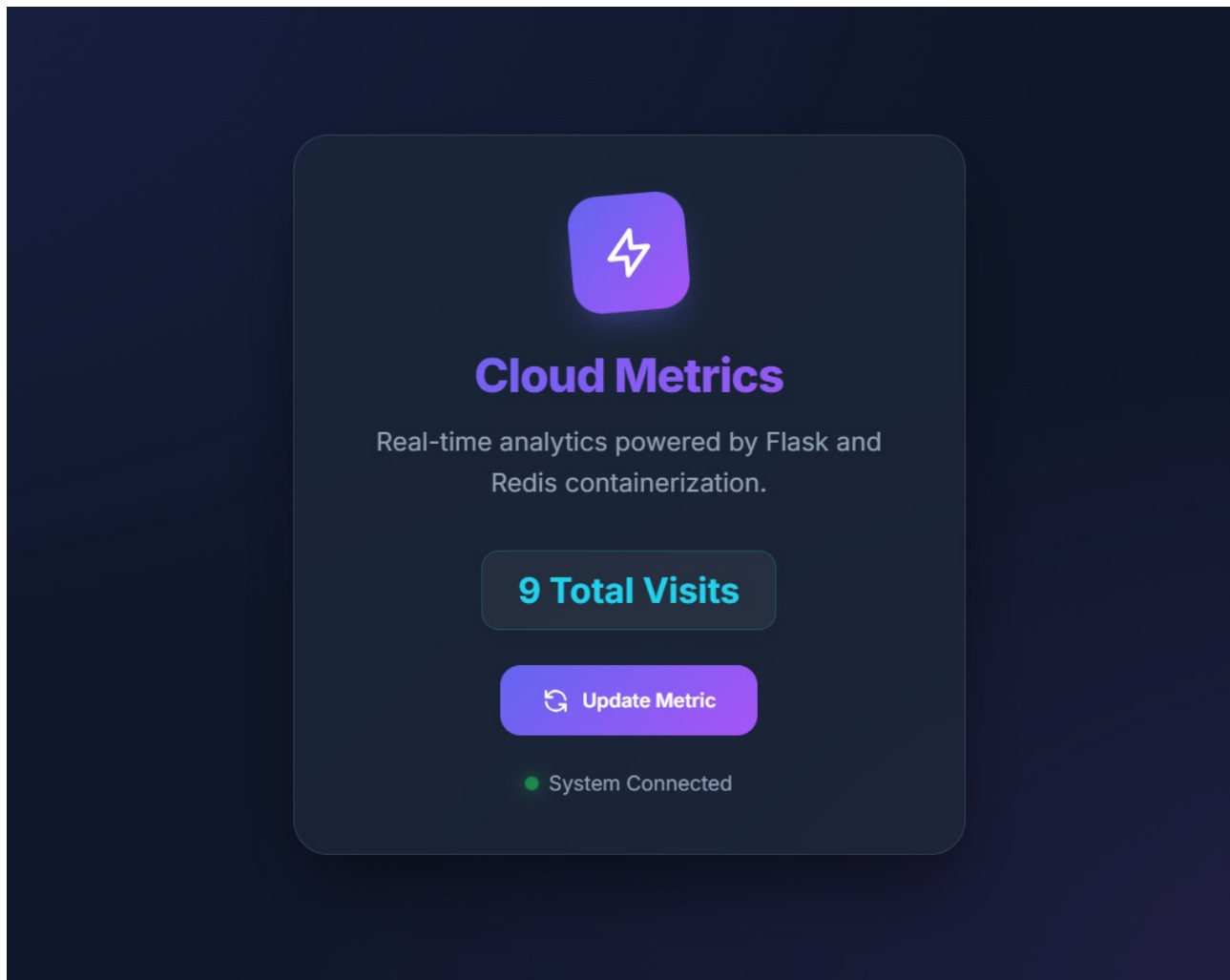


A professional, containerized microservices application featuring a **Flask** backend, a persistent **Redis** database, and an **Nginx** reverse proxy. This project demonstrates high-availability architecture, internal networking security, and cloud-ready deployment patterns.

Features

-  **Premium UI:** Modern glassmorphism design with dark mode, Inter typography, and micro-animations.
-  **Nginx Reverse Proxy:** High-performance entry point on port 80 for improved security and request handling.
-  **Secure Networking:** Redis is isolated within an internal backend network, inaccessible from the host.
-  **Data Persistence:** Named Docker volumes ensure your visit counter data survives container restarts.
-  **Health Monitoring:** Integrated health checks for Redis to ensure system reliability.
-  **Structured Logging:** Centralized Python logging for real-time monitoring of app status.

Application Interface



Tech Stack

- **Backend:** Python 3.9 (Flask)
- **Database:** Redis (Alpine)
- **Proxy:** Nginx (Alpine)
- **Containerization:** Docker & Docker Compose
- **Design:** CSS3 (Glassmorphism, Animations)

Quick Start

1. Prerequisites

- **Docker Desktop** (<https://www.docker.com/products/docker-desktop/>) installed and running.

2. Launch the Application

Clone the repository and run the following command in the project root:

```
docker-compose up -d --build
```

3. Access the App

- **Web App:** <http://localhost> (<http://localhost>) (via Nginx)
- **Direct Backend:** <http://localhost:5000> (<http://localhost:5000>)

Cloud Deployment

Repositories

All repositories within the ghassenellili123 namespace.

Q Search by repository name

All content

Create a repository

Name	Last Pushed ↑	Contains	Visibility	Scout
ghassenellili123/flask-redis-app	less than a minute ago	IMAGE	Public	Inactive
ghassenellili123/gourmet-go	5 days ago	IMAGE	Public	Inactive
ghassenellili123/getting-started	24 days ago	IMAGE	Public	Inactive
ghassenellili123/portfolio	4 months ago	IMAGE	Private	Inactive

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Docker Hub

The latest image is available on Docker Hub:

```
docker pull ghassenellili123/flask-redis-app:latest
```

AWS Elastic Beanstalk

To deploy to AWS:

1. Initialize EB: `eb init -p docker flask-redis-app`
2. Create environment: `eb create flask-redis-env`

Project Structure

├── imagedocker.png	# 3D Architecture Diagram
├── imageUI.png	# App Interface Preview
├── imageDockerHb.png	# Docker Hub Repository Mockup
├── app.py	# Flask logic & logging
├── Dockerfile	# Web container instructions
├── docker-compose.yml	# Multi-container orchestration
├── nginx.conf	# Reverse proxy configuration
├── requirements.txt	# Python dependencies
├── static/	
│ └── index.css	# Premium Design System
└── templates/	
└── index.html	# Glassmorphism Template

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Created with  for the Cloud Computing & Virtualization Mini-Project.